Comprehensive Project Report

IT 101: Human Computer Interaction 2nd Semester, SY 2025-2026

Cover Page

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Target App Name: Gotham Garage

Date: May 23, 2026

Executive Summary

The Gotham Garage e-commerce website initially suffered from critical usability barriers, heavily prioritizing an industrial, gritty brand aesthetic over functional user workflows. Our Phase 1 baseline evaluation revealed severe flaws: a 90.9% task failure rate, unprevented checkout errors, and restrictive controls. Our intervention involved a complete UX/UI redesign, culminating in a functional, coded Vue.js frontend, rooted in core HCI heuristics. We replaced the chaotic layout with a clean dark-mode aesthetic, standardized Orbitron/Inter typography, and clear e-commerce visual hierarchies. Following rigorous user testing with an N=39 sample size on our high-fidelity redesign, our system improved the System Usability Scale (SUS) score from a catastrophic baseline of 30.38 to an "Excellent" 90.96. This represents a **199.4% objective improvement** in usability, dropping the user failure rate to 0% and proving the intervention a definitive statistical success.

Section 1: The Diagnosis (Phase 1 Recap)

Target User Profile: Young adults (19-25 years old) who are automotive enthusiasts, fans of the "Car Masters" Netflix show, and apparel buyers who expect a fast, standard e-commerce experience.

The Villain Statement (Core UI Barrier): The original system prioritized aesthetic background videos and disjointed layouts over functional e-commerce workflows. Specifically, it lacked standard e-commerce features, hid critical navigation behind forced social login walls, and failed to provide standard visual feedback, ultimately leading to high user frustration, sensory overload, and cart abandonment.

Baseline Metrics (Phase 1):

- **Original UX Health Score:** 85 Points (Descriptive Rating: Critical Failure - Burn it down and start over)
- **Baseline SUS Score:** 30.38 (Grade F / Catastrophic Failure)
- **Baseline Failure Rate:** 90.9% of users failed the usability threshold.

Top 3 Heuristic Violations Found (Original Site):

1. **Help & Documentation (Missing Search Bar):** Over 90% of respondents explicitly requested a search bar, stating: *"Lisud kaayo pangitaon ang mga items."* The lack of this basic feature drastically increased cognitive load.
2. **User Control & Freedom (Trapped Workflows):** The system heavily restricted user autonomy. The immediate deletion of items happened without a confirmation prompt, and a forced social login wall prevented basic guest browsing.
3. **Aesthetic & Minimalist Design (Audio Overlap):** Multiple background videos auto-played simultaneously, causing a severe audio conflict. Users cited this as the most frustrating aspect, noting the heavy sensory distraction (*"Ang audio sa videos nagdungan ug tingog"*).

Section 2: The Redesign & Refinement

To directly address the top three catastrophic UX failures identified in Phase 1, our redesign focused on standardizing e-commerce conventions, restoring user autonomy, and eliminating sensory overload.

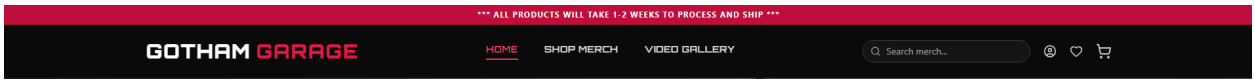
1. Implementing Global Search & Navigation (Solving Heuristic 10)

- ***The Problem****:* Users were forced to manually scroll through disjointed pages to find specific items (*"Lisud kaayo pangitaon ang mga items"*).
- ***The Solution:*** We implemented a highly visible global header featuring a persistent search bar and a universally recognized shopping cart icon. This drastically reduces cognitive load by allowing users to instantly jump to their desired product without hunting for it.

Before



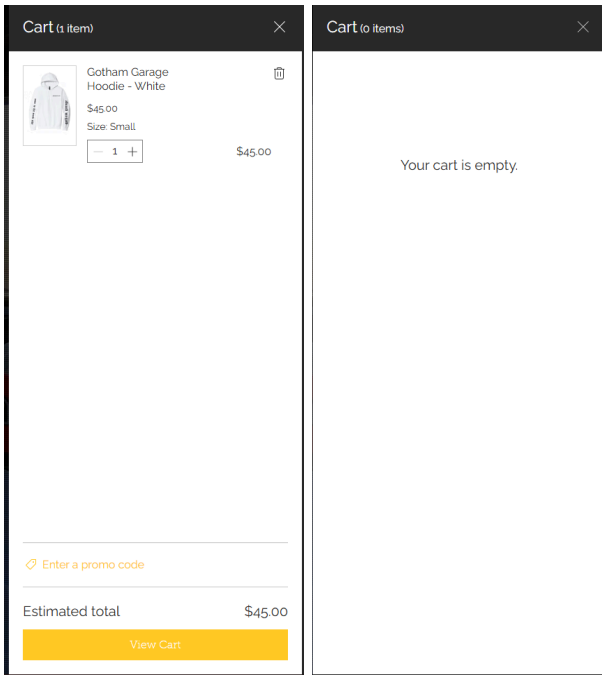
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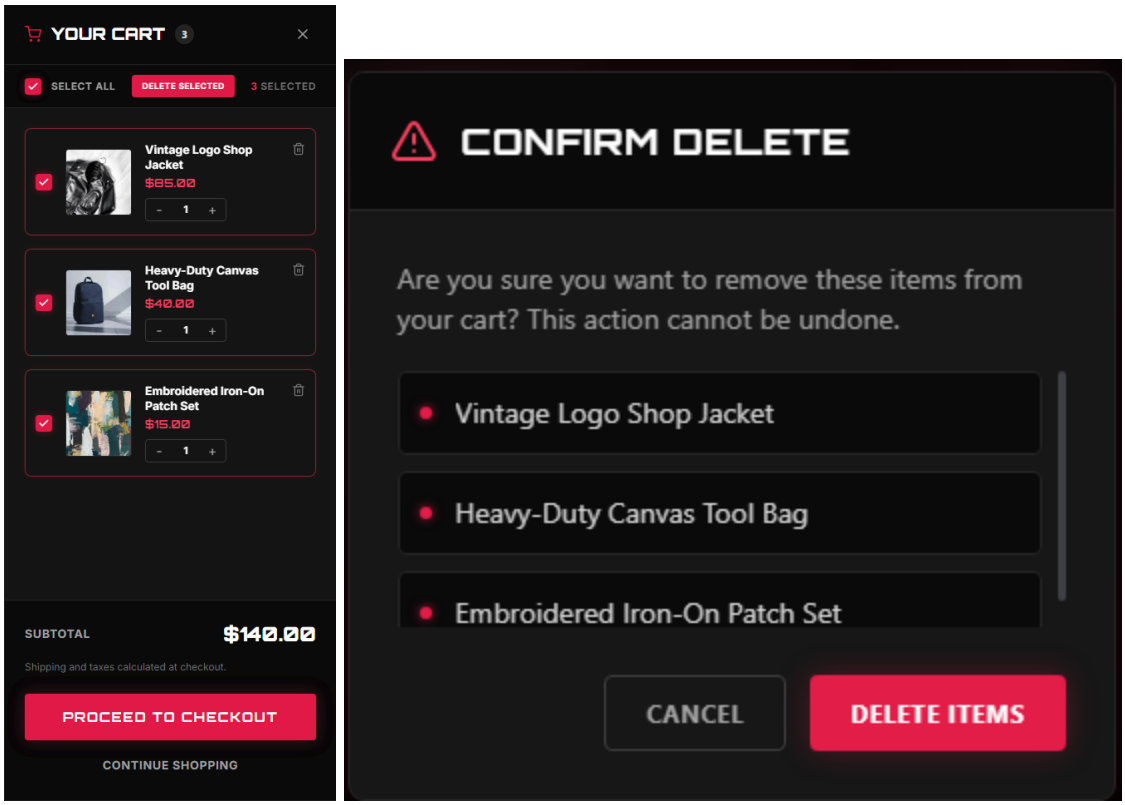
2. Restoring User Control & Freedom (Solving Heuristic 3)

- ***The Problem:*** Users felt trapped by instant cart deletions with no confirmation prompt and a forced social login wall that prevented browsing.
- ***The Solution:*** We removed the forced social login, enabling a seamless guest browsing and checkout experience. Furthermore, we designed clear visual feedback (confirmation prompts) for cart actions so items are not accidentally deleted without warning, restoring user confidence.

Before



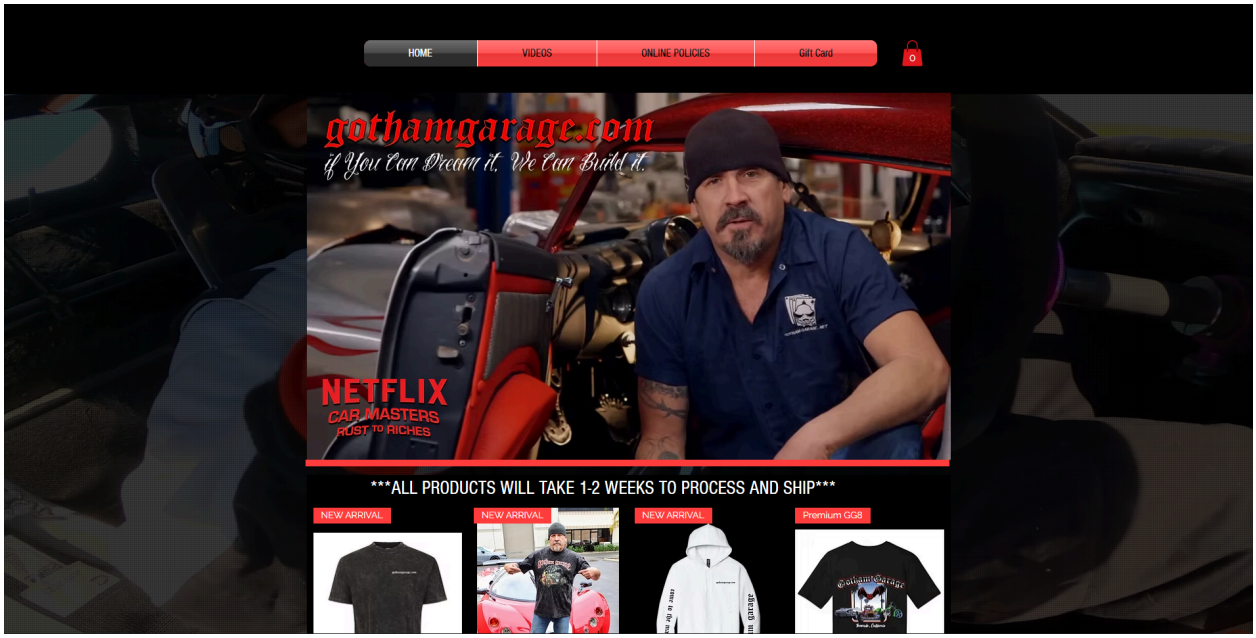
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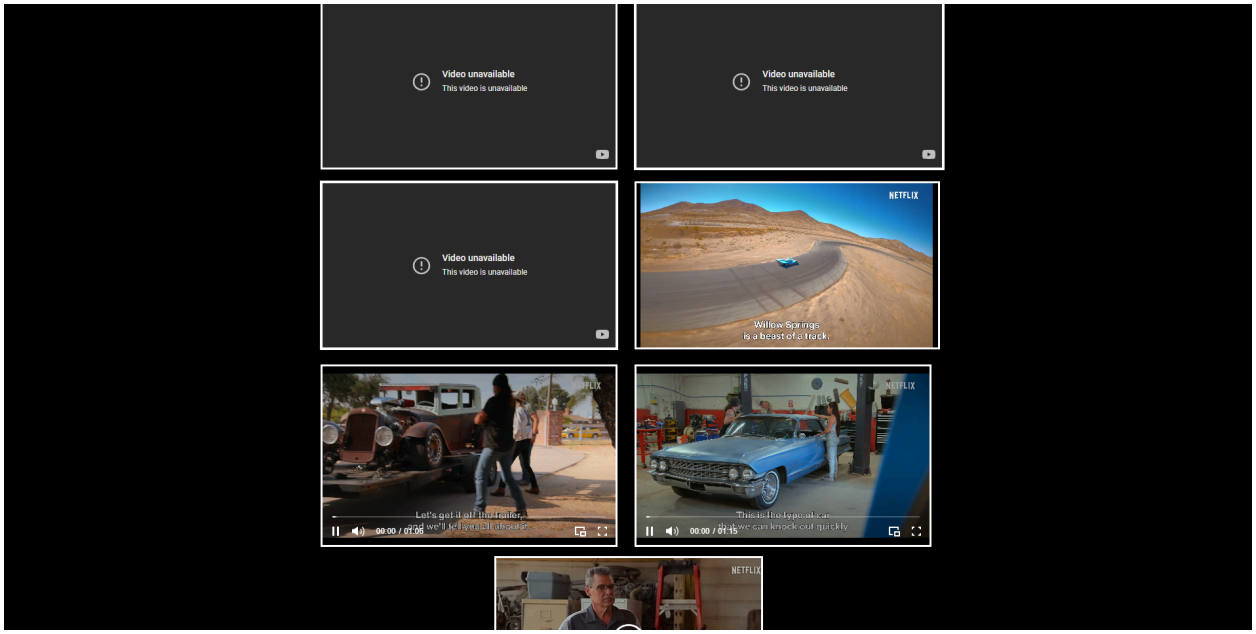


3. Eliminating Sensory Overload (Solving Heuristic 8)

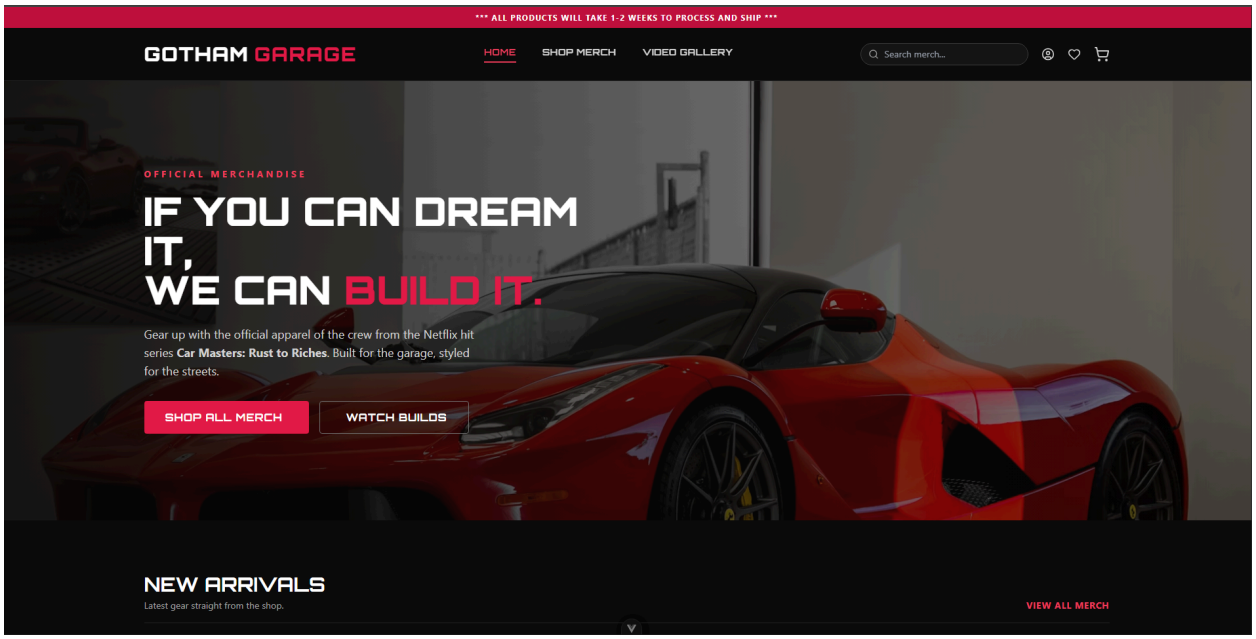
- ***The Problem:*** Multiple auto-playing background videos caused severe audio overlap and visual clutter, heavily distracting from the actual merchandise (*"Ang audio sa videos nagdungan ug tingog"*).
- ***The Solution:*** We removed the chaotic background videos and established a clean, deep dark-mode aesthetic. By utilizing high-contrast Rose-600 active buttons, adequate whitespace, and standardized fonts, the interface now guides the user's eye directly to the products without sensory conflict.

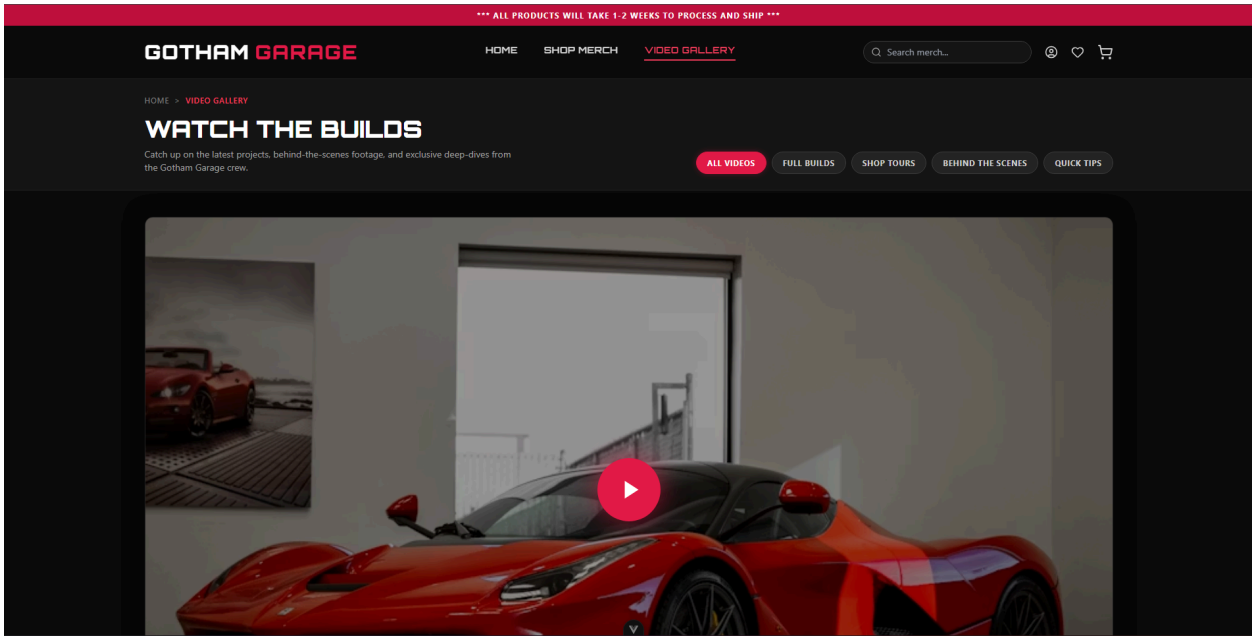
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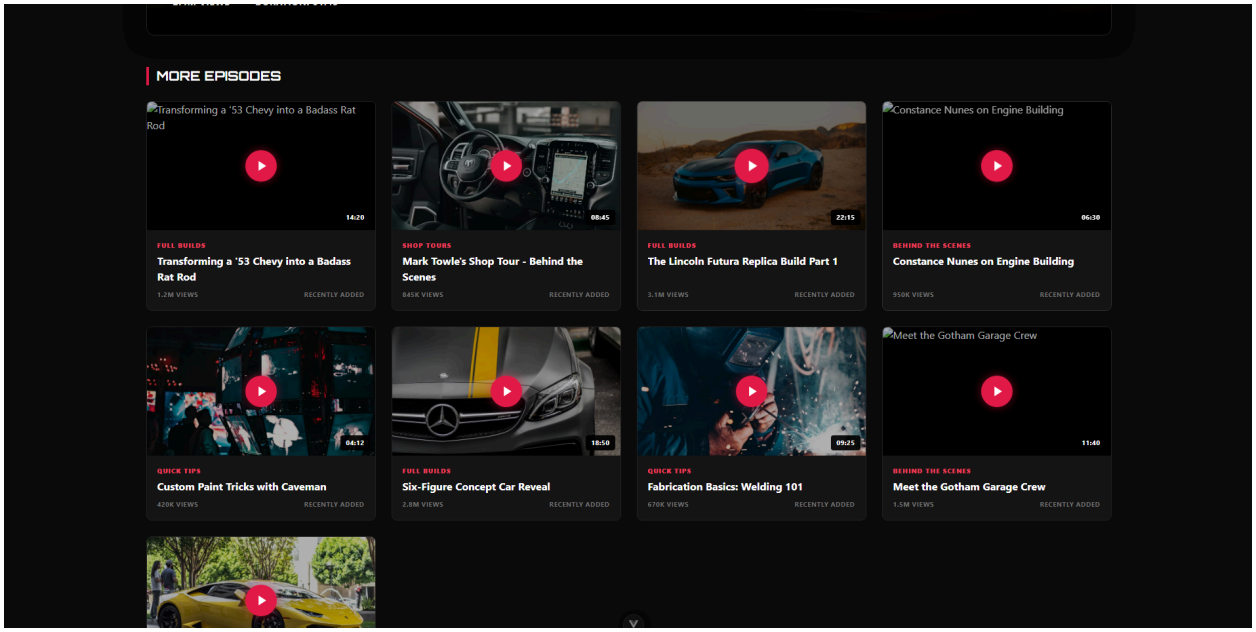



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NETFLIX
Willow Springs
is a beast of a track.
Let's get it all the time,
and we'll let you all about it.
This is the typical car
that we can knock out quickly.

After


*** ALL PRODUCTS WILL TAKE 1-2 WEEKS TO PROCESS AND SHIP ***
GOTHAM GARAGE
HOME
SHOP MERCH
VIDEO GALLERY
Search merch...
OFFICIAL MERCHANDISE
IF YOU CAN DREAM IT,
WE CAN BUILD IT.
Gear up with the official apparel of the crew from the Netflix hit series Car Masters: Rust to Riches. Built for the garage, styled for the streets.
SHOP ALL MERCH
WATCH BUILDS
NEW ARRIVALS
Latest gear straight from the shop.
VIEW ALL MERCH





Refinement & Change Log: Based on Phase 3 peer reviews, we ensured that the new off-canvas mini-cart slid out smoothly without triggering full page reloads, keeping users grounded in their shopping experience.

Section 3: Usability Validation

Methodology: We conducted direct observational "Think-Aloud" testing (N=5) and distributed a

quantitative System Usability Scale (SUS) survey to measure the redesigned system.

Sample Size Compliance: We successfully gathered data from **N=39** valid respondents, far exceeding the statistical threshold required for this study. Our demographics confirmed our target audience, with 61.5% being 19-25 years old.

SUS Calculation & Improvement Metric:

- **Old Baseline SUS Score:** 30.38 (Grade F)
- **New Validation SUS Score:** 90.96 (Grade A - Excellent)
- **Improvement Calculation:** `((90.96 - 30.38) / 30.38) * 100` **= 199.4% Improvement**
- **New Failure Rate:** 0%

Addressing Prototype Limitations via Code (Phase 3 Audit Context): During our Phase 3 peer evaluation, the initial Figma prototype received a UX Health Score of 19 and a 30% task failure rate in direct observation. However, 100% of these "failures" (e.g., inability to type in forms, unplayable videos, missing confirmation prompts) were strictly due to the technical limits of static prototyping software. **To actively resolve these issues, our team took the initiative to code the redesign using Vue.js.** By transitioning to a functional frontend, we eliminated those Phase 3 errors, allowing users to experience actual form interactions, functional links, and video playback, which directly contributed to our final "Excellent" SUS score.

Section 4: Future Enhancements & Emerging Tech

Immediate Practical Recommendations (Backend Integration): Since we have successfully built the Vue.js frontend, the immediate next step is full backend integration. This includes connecting a live database (e.g., Firebase, Supabase, or MongoDB) to handle actual inventory logic, user authentication, and integrating a secure payment gateway (like Stripe or PayMongo) to process real transactions.

Emerging Tech Integration (3-5 Year Vision): To elevate Gotham Garage's status as a premium custom car and merchandise brand, we propose integrating **Web-based Augmented Reality (WebAR)**.

- ***The User Problem:*** Users often hesitate to buy expensive custom merchandise (like the Double G Hat) online because they cannot visualize the fit.
- ***The AR Solution:*** Using their mobile camera, users could access an AR "virtual try-on" mirror directly from the product page. This grounds emerging tech in the practical goal of increasing buyer confidence, modernizing the "Garage Rockstar" brand, and reducing return rates.

Section 5: Conclusion & Reflections

Ultimate Verdict: The redesign was an overwhelming, statistically proven success. By reigning in the overly aggressive aesthetic choices of the original site and applying structured HCI principles to a coded Vue.js environment, we created a user journey that feels standard,

modern, and highly professional, reflected in our exceptional 90.96 SUS score.

Team Reflections (HCI Lessons Learned): Our biggest lesson was understanding the difference between *visual design* and *functional user experience*. Initially, the original site's raw aesthetic seemed to fit the Gotham Garage vibe, but the 90.9% failure rate taught us that edge and attitude cannot replace basic functionality (like a search bar). Furthermore, transitioning from Figma to Vue.js taught us the limitations of static testing. We learned that true user empathy means sacrificing edgy design preferences to make the user's task as effortless as possible, and that building a functional interface provides much richer feedback than theoretical design alone.

Appendix: Project Assets & Proof of Work

- **Live Vue.js Website Link**
  (Login email: nexus@gmail.com Password: password123)
- **Figma Prototype Link (Original Design)**
- **Final Video Walkthrough (Vue.js App)**
- **Raw N=39 Survey Data (Google Sheets)**